
The Beech Forests of Wisconsin--Their Phytosociology and Relationships to Forests of the State without Beech

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THE BEECH FORESTS OF WISCONSIN—THEIR PHYTOSOCIOLOGY AND RELATIONSHIPS TO FORESTS OF THE STATE WITHOUT BEECH¹

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INTRODUCTION

As a part of the intensive investigation of Wisconsin plant communities being carried out at the University of Wisconsin a study of those woods of the state in which beech (*Fagus grandifolia* Ehrh.) is a component was made. Most of the field work for this research was done during the summers of 1952 and 1953. An earlier paper (Ward 1956) dealt with compositional changes within the beech forests and the nature of the beech border in the state. The present paper is

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concerned with a third major phase of the study, the general phytosociology of the beech forests and the relationship of these to forests of the state without beech.

Data from fifty-one beech stands distributed over the beech range (Fig. 1) were included in the analysis, and data from eighty-three nonbeech stands studied by previous students at the University of Wisconsin were utilized for comparative purposes.

In selecting beech stands suitable for this study three requisites as to condition were adhered to: the presence of beech; lack of recent general disturbance of any kind; and maturation beyond a pole-tree condition if a second growth stand. Some of the stands were located by reference from



FIG. 1. The range of beech in Wisconsin and the location of beech stands used in this study.

sources such as county agents, but most were found by driving over secondary roads and investigating any woods that appeared to have possibilities. Approximately four stands were checked out for every one chosen, with cutting, and grazing by dairy cattle the primary causes for rejection.

DESCRIPTION OF THE AREA

Climate

Certain generalizations can be made about the climate of Wisconsin. Consistent with its continental location, winter and summer are distinct seasons. The winters are characteristically rather severe and snowy, particularly in the northern two-thirds of the state, but the more variable southern portion sometimes enjoys milder winters than areas further to the south. In summer the northern portion is dependably cool, whereas the southern portion experiences both times of high temperatures and high humidity, and other periods of pleasant moderation. Over the entire state summer precipitation is notably reliable, although minor droughts occur occasionally in the southern portion.

The frost-free season varies from less than 100 days per year in the extreme northern sections to as much as 175 days in southeastern Wisconsin. Snowfall is heavier in the north, averaging near 55 inches as against 30 inches in the south; total precipitation, however, is usually somewhat greater in the south. Possible sunshine for the state as a whole averages around 50 per cent.

The climatic pattern within the beech forest range, although not sharply distinct from that of other parts of the state, does show certain differences which may be critical in the limitation of beech. Areas to the northwest are on the average noticeably colder in winter, with also a substantial reduction in the length of the growing season. To the southwest of the beech forest area the somewhat higher summer temperatures may result in a significant reduction of effective moisture, the amount of actual precipitation being roughly the same in the beech forest range as in areas immediately adjacent.

Glacial History

The area of the present study was in large part covered with moraine of the Valdres stage, the last major advance of the ice sheet in eastern Wisconsin. Some of the most pronounced topographical changes of the region occur in the interlobate moraine formed by deposition between the Green Bay and Lake Michigan Lobes of the Cary advance. Called the "Kettle Moraine" because of deep hollows, it is most strongly developed in the southern portion of the beech range. The later Valdres ice advance, in all probability part of an ice sheet from the Labradorian center, was also lobate near its borders, but did not form a pronounced interlobate moraine.

The glacial deposits of eastern Wisconsin contain imported igneous and metamorphic rocks in addition to local limestone, shale and sandstone. The ground moraine which covers nearly all of the eastern part of the state is composed primarily of till, but has areas of stratified sand and gravel. The topography consists of rolling or hilly moraines and till plains, with many areas either covered by lakes or too swampy for agriculture.

Overlying the Cary drift in an area extending northward from Milwaukee and spreading westward across Calumet and Manitowoc counties is a layer of red clay presumed to be the deposit of a lake, or lakes, existent sometime between the Cary and the final Valdres advance. Some of this material was ridged up into terminal moraines by the Valdres ice. Other lake deposits are present along the shores of Lake Michigan as the result of post-glacial enlargements and recessions regulated by changes in drainage of the Great Lakes (Martin 1932).

Soils

The original soil of the beech forest area was derived from the weathering of the bedrock dolomite and sandstone. These residual soils were added to and mixed by the numerous glacial advances. In some areas lake deposits were laid

down over which the glaciers readvanced, working and shifting the lacustrine and previously mixed deposits.

The soils of the beech forest are of two basic groups, gray-brown podzolic over the lower two-thirds of the range, and podzolic in the northern counties. Within these groups several series are represented, the general distribution of which is given by Muckenhirn and Dahlstrand (1947). In the southwestern portion of the beech range where yellowish or grayish-brown loams of the Miami series predominate, the acid surface soils are a loose, friable loam to a depth of about eight inches. The brown subsoil is distinctly heavier in texture, but has a nutlike structure favorable to penetration by water and roots. The Kewaunee soils found along Lake Michigan and extending over the central portion of the beech area are very similar to the Miami, but with a more reddish color and a somewhat heavier subsoil.

Although designated podzols, classical profiles of this group are not abundant in northeastern Wisconsin, primarily because of the highly calcareous parent materials and glacial deposits which tend to inhibit acidic leaching associated with such development. Sandy areas dominated by conifers show greatest evidence of podzolization.

In contrast to the mull humus of the Miami-Kewaunee region, a fairly distinct organic layer of the mor type is usually present in the northeastern Wisconsin beech woods. Underlying this is a layer of grayish-brown silt or fine sand ranging up to about one foot in thickness, and a somewhat heavier subsoil which gives way to a highly calcareous, loose, gravelly, glacial till. Shallow, brownish-gray loam of the Longrie series, deeper clay loams of the Onaway series, and the sandy, low fertility soils of the Vilas series are most important in this portion of the beech range.

LITERATURE REVIEW

Ecological reference of various kinds is made to the beech of Wisconsin in early writings (Lapham 1854, Chamberlain 1877, Roth 1898, Brunken 1900, 1902), but there has not been a general study either descriptive or quantitative of the beech forests of this state. Only two previous studies present quantitative information on the beech forests of the state; Brown and Curtis (1952) included a few stands having beech in their study of the northern hardwoods of the state, and Gilbert and Curtis (1953) present herb data from upland forests of the state, some of which were obtained in stands that have beech.

Perhaps most closely related to the Wisconsin beech forests from a floristic standpoint are similar forests of Michigan. Studies by Quick (1923)

and Cain (1935) present data which can be used to show considerable agreement with the woods of the southern part of the Wisconsin beech range. Several species are found in the southern Michigan beech woods which are not found in the Wisconsin woods, including *Liriodendron tulipifera* L., *Nyssa sylvatica* Marsh., *Platanus occidentalis* L., *Quercus bicolor* Willd., and *Sassafras variifolium* (Salisb.) Ktze., but they make up only a small part of the tree population. Data from the northern part of the lower peninsula of Michigan (Quick 1923, Gleason 1924, Wollett and Sigler 1928) indicate a close affinity between the sugar maple-beech-conifer forests of that area and those in northeastern Wisconsin.

Studies in Indiana (Esen 1932, Potzger and Friesner 1943, Jones 1952, and Trotter 1952) provide data which also can be used to show a close relationship between the sugar maple-beech dominated stands of that state and those of this area. Here there are approximately 10 tree species which are not to be found in the Wisconsin woods, but for the most part they are of secondary importance. Williams (1936), in a biotic study of the Chagrin Reservoir near Cleveland, provides data which with little change could be taken from Wisconsin stands; there is a striking similarity in the tree, shrub, and herb composition. Data from eastern New York (Russell 1955) also indicate a basic similarity between the sugar maple-beech forests of that region and those of Wisconsin.

With regard to the taxonomic status of the beech in Wisconsin Camp had this to say (1953, personal communication):

"In Wisconsin there is no one of the basic biotypes present, rather the beech population in Wisconsin is paced by a series of genes derived from the Red Beech; however, there is ample evidence that genes derived from both the southern White Beech and the northern Gray Beech also are present in greater or less amount." Herbarium specimens annotated by Camp, and minor leaf studies by the author indicate that the variation within the Wisconsin beeches follows no particular geographical pattern, and to the present no ecological correlations have been found.

METHODS

To sample the tree population the random pairs method was employed, using a four inch or greater d.b.h. criterion for trees. In most cases forty points were taken in a woods, but in a few areas where no large stands were found, smaller woods were sampled to fill in distributional gaps.

Of the 134 stands involved in this study approximately one-sixth of them were sampled with

less than 40 points, usually 20. An estimated minimum size of the smallest stands would be between five and ten acres. The statistical inadequacy of these small samples is recognized, particularly with regard to the infrequent species, and this fact should be borne in mind by the reader. It is felt, however, that the results can be meaningfully interpreted on the basis of general trends.

To check the homogeneity of the tree population within the individual stands a chi-square test was applied. Where 40 points were taken each sheet of 10 points was used as a unit, and the observed and expected number of trees of the species with the greatest density was tabulated in each of the four units. All stands had values for "P" greater than five per cent, with most of them much higher.

Saplings (1"-4" d.b.h.) were sampled by recording by species the four nearest to the point, or in some instances the number less than four that were within a distance of twenty feet from the point. At alternate points (or at every point where only twenty points were taken) a one-meter square quadrat was laid down to sample the herb, shrub, and seedling populations. Stand presence was noted for all ferns and spermatophytes.

In order to obtain a picture of the changing herb pattern as the growing season progressed, several stands were sampled for early spring herbs in addition to the regular sample taken after the tree leaves had reached full development.

The data were used to compute an importance value (summation of relative values for frequency, density, dominance) for each tree species in a stand, and data from the lower strata were used to compute relative density for saplings, and frequencies for herbs, shrubs, and tree seedlings.

Although the study of the environmental factors was not emphasized, some data were gathered. Soil pits were dug in most stands, and notes on the profile were recorded. Three separated samples of the A₁ horizon were taken in a woods, at places not decidedly wet or dry, and combined into a single sample for laboratory determination of moisture holding capacity, and for chemical analysis.

Using a Weston photometer, light readings were taken in eighteen stands, after the trees had fully leafed out. This was made as objective as possible by pacing an arbitrary constant distance, placing the meter on the ground, and putting the light target immediately above the head while kneeling to read the dial; visual bias was largely eliminated in this way.

RESULTS

In order to analyze the unity of the beech forests and to relate them to the southern upland hardwood and northern conifer-hardwood forests of the state, it was decided to place the individual beech stands into northern and southern categories, using primarily the presence or absence of eastern hemlock (*Tsuga canadensis* (L.) Carr.) and yellow birch (*Betula lutea* Michx.) as the basis. Beech stands having one or both of these species were with one exception considered to be northern; a few stands lacking these were placed in the northern group according to other species. A summary of the two types follows.

Southern Beech Stands

Only sugar maple (*Acer saccharum* Marsh.) had an importance value in all stands with beech. *Ostrya* (*Ostrya virginiana* (Mill.) K. Koch.) and basswood (*Tilia americana* L.) were sampled in all but one, and white ash (*Fraxinus americana* L.) and red elm (*Ulmus rubra* Muhl.) were found in approximately three stands out of four. The majority of species, however, were sampled in less than half of the stands, with several found in only one or two. The decided superiority of sugar maple and beech is more readily seen in a tabulation of the average importance values for each species in all stands (Table I). Together these

TABLE I. Tabulation of tree composition of the "southern" beech forests

	Number of stands of occurrence	Average importance value	Maximum importance value
<i>Acer rubrum</i>	1	0.2	3.4
<i>Acer saccharum</i>	17	107.6	234.5
<i>Betula lutea</i>	1	0.2	2.9
<i>Betula papyrifera</i>	3	2.3	25.0
<i>Carya cordiformis</i>	4	0.8	4.4
<i>Carya ovata</i>	4	4.5	39.0
<i>Crataegus</i> spp.....	1	0.2	6.0
<i>Fagus grandifolia</i>	17	78.2	161.0
<i>Fraxinus americana</i>	13	11.5	27.7
<i>Fraxinus nigra</i>	4	1.5	13.6
<i>Ostrya virginiana</i>	16	19.3	65.9
<i>Populus grandidentata</i> ..	2	0.9	8.1
<i>Prunus serotina</i>	8	4.0	13.0
<i>Prunus virginiana</i>	1	0.2	3.4
<i>Quercus alba</i>	7	3.8	20.0
<i>Quercus macrocarpa</i>	2	0.5	5.0
<i>Quercus rubra</i>	9	15.1	78.0
<i>Tilia americana</i>	16	28.8	69.1
<i>Ulmus americana</i>	6	6.2	49.0
<i>Ulmus rubra</i>	12	13.8	95.0

two species make up more than half of the total possible value of 300. Other species show fairly high importance values in certain stands, but the averages for all stands in no case approached that of these two climax species. Basswood, next be-

low beech in importance had an average importance value considerably less than that for beech. *Ostrya*, although usually found with sugar maple and beech, seldom has a high importance value. White ash and red elm, common constituents of the southern beech forests of the state, are seldom important. Red oak (*Quercus rubra* L.) is sometimes important in beech stands, but occurs usually as a large tree and is insignificant in the small size classes.

Sapling reproduction consists mostly of sugar maple, beech, and *Ostrya*, with the former often very abundant. Sugar maple is definitely superior in seedling reproduction. The dominance of sugar maple and beech is further seen in the leaf litter, where leaves of other species are much outnumbered.

In early spring the forest floor is often covered by a few species of herbs which complete much of their life cycle prior to full foliage development of the trees; normally to be found with these are other species which carry over into the summer. Table II lists the average and maximum frequencies in seven stands for several of the most important herb species found in spring. The strictly prevernal species quite often show high frequencies in one or two stands, while being rare in others.

By mid June the appearance of the forest floor is quite different. Such species as *Allium tricoccum* Ait, *Claytonia virginica* L., and *Erythronium albidum* Nutt. have completed their life cycle, and the per cent cover by herbs is considerably less. Average and maximum frequencies for important herb species found in summer in some of the southern beech stands are given in Table II.

Northern Beech Stands

In the northern beech forests, sugar maple again was the only species with an importance value in all stands with beech (Table III). The next most important species, hemlock, *Ostrya*, yellow birch, and white ash, all were sampled in less than two-thirds of the stands, and half of the species had importance values in three or less of the 29 stands. The average importance values show sugar maple and beech to be about equal, making up approximately two-thirds of the possible value. Hemlock, although next in average importance, makes up only about one-tenth of the total value of 300, and yellow birch has a considerably lower value than hemlock. Occasionally beech occurs in stands dominated by hemlock or white cedar, but usually where beech is sufficiently abundant in a stand to be recorded at a point, it along with sugar maple dominates the tree stratum.

Sapling and seedling reproduction is usually

TABLE II. Some of the most important herbs found in representative southern beech stands. The stands sampled in summer were in some cases different from those used in the spring

	SPRING		SUMMER	
	Average frequency	Maximum frequency	Average frequency	Maximum frequency
<i>Allium tricoccum</i>	25.0	80.0		
<i>Carex pensylvanica</i>	18.6	40.0		
<i>Circaea quadrivulcata</i>			24.6	75.0
<i>Claytonia virginica</i>	23.6	100.0		
<i>Erythronium albidum</i>	11.4	80.0		
<i>Galium aparine</i>			30.0	100.0
<i>Galium concinnum</i>	16.4	50.0		
<i>Geranium maculatum</i>	22.8	45.0	17.5	45.0
<i>Hepatica acutiloba</i>	20.7	85.0	28.3	60.0
<i>Hepatica americana</i>	13.6	80.0		
<i>Osmorhiza claytoni</i>	16.4	50.0	37.5	80.0
<i>Podophyllum peltatum</i>	22.8	45.0		
<i>Polygonatum pubescens</i>	36.4	65.0	30.8	60.0
<i>Sanguinaria canaensis</i>	15.7	50.0	30.4	70.0
<i>Smilacina racemosa</i>			17.5	45.0
<i>Trillium grandiflorum</i>	36.4	65.0	18.7	60.0
<i>Viola cucullata</i>			16.7	50.0

TABLE III. Tabulation of tree composition of the "northern" beech forests

	Number of stands of occurrence	Average importance value	Maximum importance value
<i>Abies balsamea</i>	3	0.6	7.0
<i>Acer rubrum</i>	10	4.5	55.2
<i>Acer saccharum</i>	29	100.4	209.8
<i>Acer spicatum</i>	1	0.1	3.0
<i>Amelanchier</i> sp.....	1	0.1	3.0
<i>Betula lutea</i>	16	10.9	67.1
<i>Betula papyrifera</i>	13	9.8	72.0
<i>Carya cordiformis</i>	2	0.5	8.8
<i>Fagus grandifolia</i>	29	96.9	244.4
<i>Fraxinus americana</i>	16	4.2	18.0
<i>Fraxinus nigra</i>	2	0.6	13.0
<i>Ostrya virginiana</i>	17	7.7	39.3
<i>Picea glauca</i>	1	0.1	3.0
<i>Pinus strobus</i>	9	6.6	52.0
<i>Populus balsamifera</i>	1	0.1	4.2
<i>Populus grandidentata</i>	3	1.0	18.0
<i>Populus tremuloides</i>	1	0.4	11.9
<i>Prunus serotina</i>	2	5.2	7.2
<i>Quercus rubra</i>	9	5.7	51.6
<i>Thuja occidentalis</i>	3	4.2	107.5
<i>Tilia americana</i>	14	5.7	25.0
<i>Tsuga canadensis</i>	18	33.7	198.3
<i>Ulmus americana</i>	6	3.6	31.8
<i>Ulmus rubra</i>	3	2.1	39.6

predominantly sugar maple, with beech less abundant but much more important than other canopy species. The herb aspect pattern is similar to that of the southern beech forests. Several species literally cover the forest floor before the canopy is closed, but die back within a period of a few weeks. Important herbs sampled in spring and in summer are listed in Table IV.

For the analysis of the relationship between the beech and non-beech forests, data from the appro-

TABLE IV. Some of the most important herbs found in representative northern beech stands. The stands sampled in summer were in some cases different from those used in the spring

	SPRING		SUMMER	
	Average frequency	Maximum frequency	Average frequency	Maximum frequency
<i>Aster macrophyllus</i>			12.3	70.0
<i>Claytonia virginica</i>	17.0	60.0		
<i>Erythronium americanum</i>	35.5	70.0		
<i>Hepatica acutiloba</i>	11.0	70.0	11.9	65.0
<i>Maianthemum canadense</i>	24.0	70.0	20.4	85.0
<i>Osmorhiza claytoni</i>	14.0	65.0	15.0	95.0
<i>Polygonatum pubescens</i>	41.5	85.0	30.4	70.0
<i>Smilacina racemosa</i>			10.8	60.0
<i>Trillium grandiflorum</i>			14.4	50.0
<i>Viola pubescens</i>			13.8	70.0

priate northern or southern beech stands were combined with data taken from 40 stands used in the study of the northern conifer-hardwood forest (Brown and Curtis 1952), or with data from 43 stands used in the study of the southern hardwood forest (Curtis and McIntosh 1951). The selection of these stands was made without knowledge of location or of composition, except as a maximum range of variation in the importance value of sugar maple was sought. The reasons for the latter will be apparent in the following discussion.

In the treatment of data from several somewhat linearly related variables—in this case tree species—it is sometimes suitable to set one aside and plot the behavior of all others against changes in the one chosen. In this way the behavior of all others with respect to a single variable can be found, and the relationships of the variables to one another is indicated. In the selection of a pivotal variable—or species—it is essential that it occur in all of the samples. The species chosen can otherwise be any one of the group, but the results are most easily interpreted if the species is either at the beginning or at the terminus of the linear arrangement.

The above approach was used to analyze the data obtained in the present study. Sugar maple was thought to be the most suitable as a pivotal species because it satisfied the requirement for constancy in the samples and had the preferred characteristic of being a climax species. Within the area of beech, however, it was felt that sugar maple alone could not be used. A low importance value of sugar maple in non-beech woods of the state usually indicates an intermediate condition; in beech woods it may or may not, depending upon the importance of beech. In a particular stand sugar maple may be low in importance while beech is quite high, the result being a relatively climax woods. Obviously the behavior of most

species, would be quite different in intermediate woods with little sugar maple and much oak from that in woods with a similar small amount of sugar maple but with much beech.

There is some indication in Wisconsin that beech is less climax than sugar maple (Ward 1956), but for the present purpose it may be considered close enough to sugar maple that in a general way the two species are interchangeable within the Wisconsin range of beech. The sum total of beech and sugar maple in a given woods is therefore assumed to represent a "climax unit" in which the two species may vary to produce a similar result. That is to say, the species-environment complex present in a woods composed of 60 per cent sugar maple and 30 per cent beech is reasonably similar to that in a woods of 30 per cent sugar maple and 60 per cent beech. It is not to be taken that the conditions and species are precisely the same, but rather it is intended that for purposes of comparison it is meaningful to relate other quite different species to this composite unit of relatively narrow limits. Outside of the range of beech, sugar maple alone was considered to be the climax unit against which other species could be plotted.

The data from each stand were treated in a similar manner. After subtracting the sum of the importance values for sugar maple and beech from the total stand value of 300, the individual importance values for the remaining species were taken as a percentage of the residual value. This method indicates the behavior of a species against a change in the climax unit without being mathematically dependent upon the change.

In plotting the results each species was taken separately. The ordinate value represented the per cent value the species was of the residual value, and the abscissa indicated the importance value of the climax unit. For most of the length of the abscissa the theoretical percentage value that one tree makes up of the residual value is low, but as the value of the climax unit approaches 300 the percentage rises abruptly. In view of the unwarranted weight for the chance occurrence of a single tree, only data below a value of 270 for the climax unit were considered, except where the point was based upon 5 or more trees in a total sample of 80.

That the combination of sugar maple and beech into a climax unit is a reasonable manipulation is borne out by a comparison of the behavior of a species in stands having more sugar maple than beech with the response of that species in stands in which the reverse is true. In Fig. 2, the behavior of basswood, as an example, is seen to be

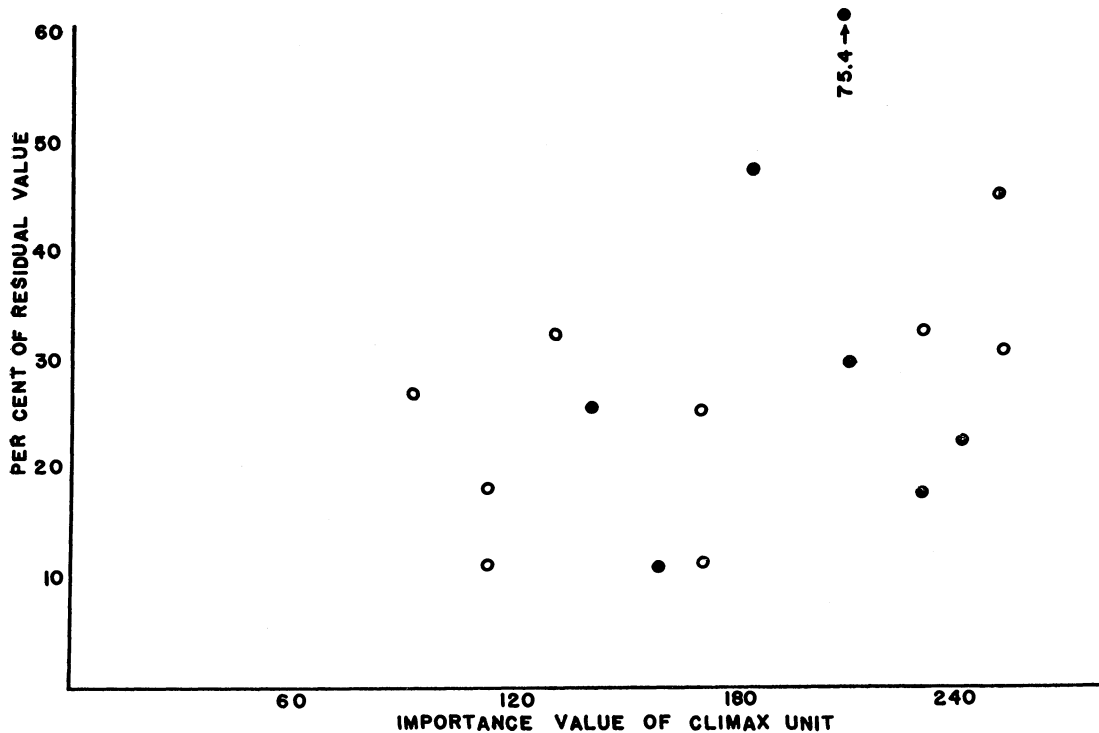


FIG. 2. The relative importance of basswood in stands having a variation in the superiority of sugar maple and beech. Circles indicate stands in which the importance value of sugar maple exceeds by at least 10 that for beech; dots indicate a similar superiority for beech. The "climax unit" is considered to be the combination of sugar maple and beech importance, and the residual value is the summation of importance values for species other than these two.

essentially independent of the varying superiority of the two dominant species.

Among the tree species that had importance values in one or more of the southern beech stands seven occurred sufficiently often to indicate a trend, while nine species were abundant enough to be used in the northern beech stands.

Importance Value Trends in Southern Beech Stands

White oak (*Quercus alba* L.) was sampled fairly often in stands in which sugar maple and beech are low in importance, but it was sampled less frequently as the climax species are more numerous. Corresponding to the drop in occurrence is a decline in importance, so that in stands with roughly 50 percent or more of the trees sugar maple and beech, white oak is of little consequence. The trend for this species indicates that the peak does not fall anywhere within the gradient of the climax unit, but occurs in more pioneer stands. There was no evidence of reproduction of this species in the beech stands.

Red oak was sampled in most of the stands that have white oak, but was also found in several stands higher up the climax unit gradient. The

importance of this species is often quite high, with several values above 25 per cent of the residual value. Its occurrence in the samples dropped quite noticeably with values of the climax unit above 130. Seedlings and saplings of red oak are rare in southern beech woods.

Red elm was sampled in a majority of stands having sugar maple and beech, and seems to peak in importance where sugar maple and beech make up slightly less than half of the trees.

American elm (*Ulmus americana* L.) seldom occurs in stands that have sugar maple and beech, being sampled slightly more often in stands with sugar maple only. Elm is exceptionally low in relative importance over most of the gradient, but a few high values in the range of 150-210 result in a definite upward trend. Beyond this the behavior is uncertain because of its absence in the more climax stands. Reproduction of both species of elm is variable, but mostly unimportant in woods having beech.

White ash (Fig. 3) has a low level of importance in those stands with small values of the climax unit, but shows a rather even rise to a medium level of importance. In no stands did it constitute over 40 per cent of the residual value.

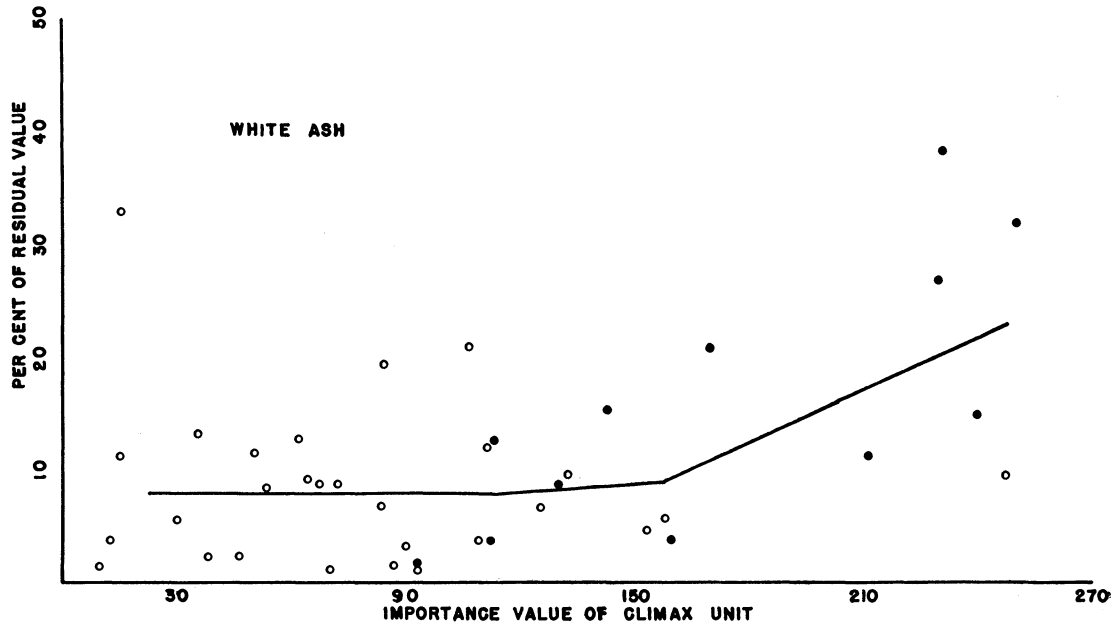


FIG. 3. The importance value trend for white ash across the non-beech-beech gradient in southern Wisconsin. Dots indicate stands that have beech; circles represent stands in which sugar maple alone constitutes the climax unit. The solid line is a running mean.

Although a fairly tolerant species, reproduction is seldom abundant in the beech woods.

Ostrya was sampled throughout the range of the climax unit, occurring in all but six of the stands, and in all but one of the stands in which the combined importance values for beech and sugar maple was greater than 150. Always below 20 per cent of the residual value in the first half of the linear increase in sugar maple and beech, several high values are found in the last half, including one stand where *Ostrya* was the only tree recorded at the points besides sugar maple and beech. Never a canopy species, *Ostrya* often makes up a large percentage of the small tree and sapling layers.

Basswood also shows high occurrence in the samples, being recorded in all but five of the stands, four of which are in the 1-30 range of the climax unit. Where sugar maple and beech occur together the chances are very great that basswood will also be found, and often in considerable amounts. The plot of the residual values indicates a gradual rise in importance as sugar maple and beech are also more important. Although found in most southern beech stands, seedlings and saplings of basswood are seldom abundant. Many of the saplings which do occur are the result of stump sprouting.

Importance Value Trends in Northern Beech Stands

In general, species other than sugar maple and beech were sampled much less frequently in the

northern woods than were similar species in the southern stands, with none approaching *Ostrya* and basswood in the south.

White pine (*Pinus strobus* L.) seldom occurs with sugar maple and beech, but is found most often where these species are low in importance. An intolerant species, it does not reproduce except in openings of sugar maple-beech stands.

White birch (*Betula papyrifera* Marsh.), with medium sampling occurrence among the northern trees, is quite infrequent in woods with more than half of the trees beech and sugar maple. A peak in the moving average is suggested near a climax unit value of 60, with several high ordinate values in this area. From here the trend is essentially downward. White birch does not reproduce in beech stands.

Red maple (*Acer rubrum* L.) with an importance value in slightly less than half of the stands, seldom makes up more than 20 per cent of the residual value. A slight peak is suggested near the center of the gradient, beyond which this species drops both in occurrence and importance. Reproduction is occasionally quite abundant, but apparently little of it survives.

A species sampled infrequently across the entire gradient is red oak. Even so there appears to be a general rise in importance corresponding with an increase in the climax species. As in the southern beech stands, only occasional seedlings and saplings are found.

The behavior of hemlock appears to be some-

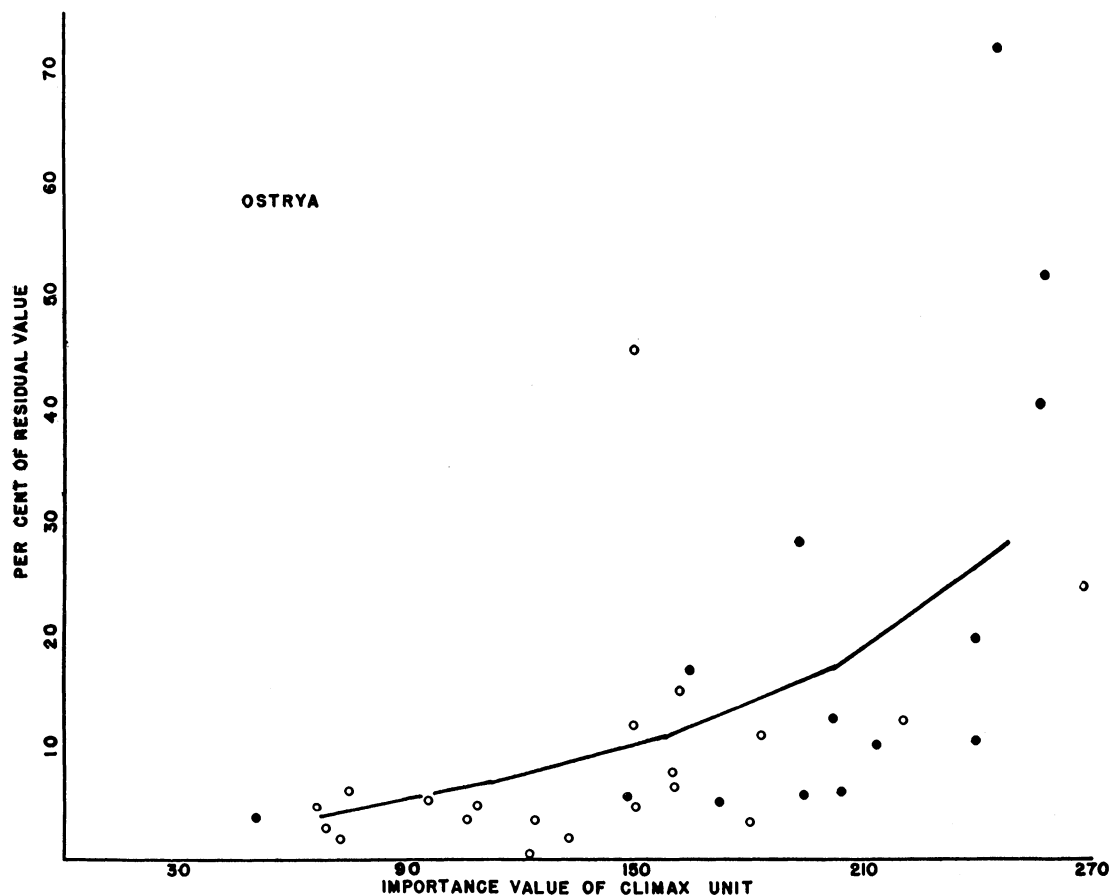


FIG. 4. The importance value trend for *Ostrya* across the non-beech-beech gradient in northern Wisconsin. Dots indicate stands that have beech; circles represent stands in which sugar maple alone constitutes the climax unit. The solid line is a running mean.

what irregular when compared with the presumably less tolerant species white birch, red maple, and red oak. The major peak for hemlock is at a lower value on the climax unit gradient than for any of these species. It should be emphasized that the linear treatment of data against an increase in sugar maple and beech does not necessarily indicate a linear relationship of all species plotted against this gradient. Hemlock probably enters the beech forest complex in different ways. In one case it may come in abundantly after some rather severe disturbance such as fire (Maissurow 1941) or windthrow, and attain very high importance. Beech and sugar maple gradually invade, and hemlock, while remaining higher than most other species in relative importance, shows a decided downward trend.

A second possibility is a regular successional sequence involving the replacement of less climax species. Neither mode of entry allows hemlock to maintain a high level of importance in the most climax stands. It reproduces poorly in its own shade and its saplings are heavily outnumbered

by those of sugar maple and beech even where these species form but a small percentage of the tree canopy.

Yellow birch is often associated with hemlock but seldom is as important. Although reproducing poorly in stands dominated by sugar maple and beech, this species was sampled more frequently than hemlock in stands with maximum amounts of the climax species.

Below a climax unit value of 60 basswood is very rare, while above this its occurrence is quite high, and its relative importance suggests a general rise to the end of the gradient. However, basswood was again sampled rather infrequently in the stands that have maximum amounts of beech and sugar maple. It appears that basswood comes in after sugar maple is established and increases along with it, but seldom maintains itself into the most climax stands. Reproduction although usually not abundant is evidently quite successful.

White ash was rarely sampled in stands that have less than 120 as a value for the climax unit,

suggesting that it also comes in only after sugar maple and beech are well established. Above this value white ash was sampled in most of the stands, although its relative importance remains low until the last few points. Of the stands with maximum amounts of sugar maple and beech, white ash is often one of the few other trees. Reproduction of this species is nowhere abundant.

The trend for *Ostrya* (Fig. 4) is very similar to that of white ash except that its relative importance in the upper half of the climax unit gradient is greater. As in the southern beech stands this species is often abundant as a sapling and a small tree, but rarely grows to a diameter greater than twelve inches.

For further evidence of the relationships within the beech forests, and between the beech forests and the other upland forests of the state, data from the herbaceous layer were compared. For this purpose stands from the southern hardwood and northern conifer-hardwood forests were selected on the basis of having amounts of sugar maple comparable to the range of the climax unit in the beech forests. This seemed to be a more meaningful comparison than a random sample of all woods in these two types.

Using the same division of southern and northern for the beech forests, the frequencies for all species in all stands of a particular forest category were averaged to give a composite stand. These data were then applied in a formula presented by Motyka, Dorbrzanski and Zawadzki (1950), which allows a comparison of stands based on quantitative measures other than presence. It was employed as follows:

$$\frac{2w}{a+b}$$

with w equal to the sum of the lesser average frequencies for species in common to the two composite stands being compared, and a the sum of all average frequencies of the first stand and b the sum of all average frequencies in the second stand. Equivalent stands would give a value of 1.00.

The values obtained are most useful on a comparative basis, because depending upon the data compared the indicated degree of affinity will vary considerably; close affinity from a relative standpoint between composite stands is indicated by a smaller value than that necessary to indicate similar affinity between single stands, for instance.

The closest correlation was found between the southern hardwood and the southern beech, with a value of 0.61 indicating a considerable amount of similarity. The value of 0.51 obtained when the southern beech and the northern beech were

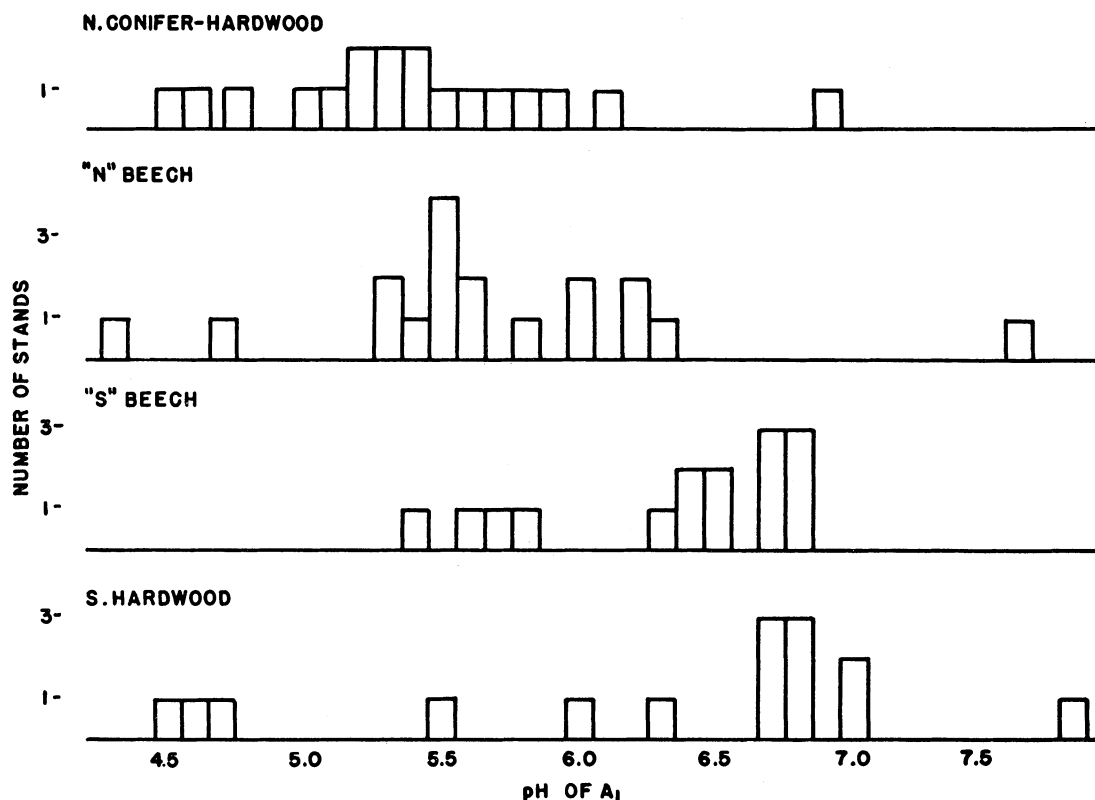
compared was notably lower. Probably not significantly different from this is the value of 0.53 for the comparison of the northern beech and the northern conifer-hardwood forests. Decidedly lower than any of these is the calculated value of 0.37 for the association of northern conifer-hardwood and southern hardwood forests. The pattern of relationship appears to be southern hardwood—southern beech—northern beech—northern conifer-hardwood, with the herb cover rather closely related progressively, but with the first and the last quite distant. To test this order a comparison was made between the southern hardwood and the northern beech forests. If the order is correct the value obtained should be somewhat less than the value for the southern beech X northern beech (0.51), and somewhat greater than the value for southern hardwood X northern conifer-hardwood (0.37). The value obtained was 0.41 which confirms the above order.

A further comparison within the beech forests was made using early spring herb data. A value of 0.46 was obtained when the southern and northern beech forests were compared. This is somewhat less than the value obtained with summer herb data, suggesting less similarity in the early part of the growing season.

Additional evidence for the relationships between the various forest categories was found in certain soil characteristics. As with the herb data stands from the northern conifer-hardwood and southern hardwood forests were selected on the basis of having amounts of sugar maple within the range of the climax unit in beech stands. The A_1 horizon, thought to be the most indicative of vegetational influence, was used in the analysis of all factors.

The pH values of the soil samples in the various forest categories are shown in Figure 5. Although the number of stands within a category is probably not sufficient to accurately show the possible extremes, the general range of the pH for the majority of the stands is indicated. The pattern is very similar to that obtained in the herb analysis. That is, southern hardwood—southern beech—northern beech—northern conifer-hardwood, with the first two most closely related and the first and the last quite distant. The lower pH values in the north are most likely related in part to the presence of hemlock with its more acid decomposition products. Moisture holding capacity data also indicate a similar relationship of forest types, with the higher values coming in the northern forests, accountable perhaps to greater amounts of undecomposed organic matter.

A further comparison within the beech forests was made using light data; such data were not

FIG. 5. pH values in the A₁ soil horizon.

available from the northern and southern upland forest studies. For this analysis it was felt that light readings in a stand could not be averaged because of the heavy influence of one or two very high readings on otherwise consistently low values. Although not widely divergent, light readings were generally lower in the northern than in the southern beech stands. The largest number of readings in the northern stands were in the intensity class of 51-100 foot candles, with over eighty per cent falling between zero and 200 foot candles. The peak in the southern stands occurred in the 101-200 intensity class, with approximately seventy per cent falling between 51 and 400 foot candles. Scattered hemlock amongst beech and sugar maple seem to result in generally low readings and this probably accounts in the main for the lower readings in the northern stands.

DISCUSSION AND CONCLUSIONS

In several recent studies in this area (Curtis and McIntosh 1951, Brown and Curtis 1952, Gilbert and Curtis 1953, Curtis 1955, Swindale and Curtis 1957) and elsewhere (Whittaker 1951, 1956) it has been found that the vegetation concerned does not segregate itself into discrete community types, but rather exists as a continuously varying gradient. The present study had as one

of its primary purposes an analysis of the evidence for or against a vegetational discontinuity related to the presence or absence of beech as a member of the forest community.

Considering first the southern Wisconsin upland forests, if Sugar Maple-Basswood and Beech-Maple are distinct types, as generally regarded (Braun 1950, Oosting 1956), there should be compositional and environmental evidence to support such a division. However, when the relative importance of secondary tree species was plotted independently against the increasing gradient of terminal species importance (sugar maple or sugar maple plus beech) no discontinuity was found between the supposed forest types. The points seemed rather to indicate a continuous trend between those woods with beech and those without. The evidence from the tree stratum was therefore taken to be decidedly in favor of an interpretation of continuity.

Although herb trends along a similar gradient were not studied, the data indicate that it would be difficult to fit all herb combinations into strict Sugar Maple-Basswood and Beech-Maple categories, a gradient again being indicated. The study by Gilbert and Curtis (1953) may be cited as evidence for continuity in the herbaceous layer

in southern Wisconsin hardwood forests. The soil characteristics pH and moisture holding capacity likewise evidence a continuity between the respective forest types, suggesting that this portion of the environmental complex is very similar in climax and near climax woods whether beech is present or not.

The presence and dominance of beech, used as a basis for the separation of associations, does not bring about a significant discontinuity when the total community is viewed. On bases other than the presence or absence of this species the Sugar Maple-Basswood and Beech-Sugar Maple in Wisconsin are not readily separable types, but rather does it appear that they are part of a larger vegetational unit in which the ranges of species vary but coincident vegetational discontinuities do not necessarily result.

The contention here is not that these two types and all of the associated developmental communities cannot be separated except on this single floristic basis. At certain compositional and environmental points away from the level of intergradation the differences might well outweigh the evidence for a gradient. The argument is rather that at the level of intergradation—the climax and near climax communities, the total evidence supports an interpretation of continuity. As designations for discrete forest associations Beech-Maple and Maple-Basswood have little validity, but the usefulness of such arbitrary designations for portions of a gradient justifies their continued usage.

With regard to the general interpretation of the beech and non-beech forested areas of the northern part of the state the problem is somewhat different. The upland forests of northern Wisconsin are generally included in a Conifer-Hardwood forest (Nichols 1935, Braun 1950, Oosting 1956) in which the absence of certain dominant species, including beech is not used for a separation of major types, but only for the separation of sub-types or sections. The general mutuality of ranges of a basic group of hardwoods and conifers and the general similarity of climax and developmental communities within the limits of the forest area are apparently considered to overshadow any disparity in the ranges of dominant species.

Despite the absence of any claim for a major division based upon the absence of beech it is again of interest to know something of the nature of the relationship between the beech and non-beech forests in this area. Is the evidence in favor of a discontinuity or a continuity? Using, as with the southern stands the evidence of secondary species trends, herb correlations, and soil and light data it appears that in the northern forests as well

as in the southern forests a vegetational continuity exists despite a floristic discontinuity involving a dominant species—beech.

A primary conclusion to be drawn at this point is that it is possible to integrate the beech forests with other upland forests of the state. In effect the continua arrangements of the southern Wisconsin hardwood forest (Curtis and McIntosh 1951) and the northern conifer-hardwood forest (Brown and Curtis 1952) may be extended to include the forests dominated by sugar maple and beech.

Of considerable interest is the possible continuity between the southern and northern beech forests of the state and a derived continuity of the southern hardwood and northern conifer-hardwood forests by way of the beech forests. Here the affinity is not so marked and the evidence is probably insufficient for any decision on continuity or discontinuity. However, it is true that the vegetational discontinuity between the northern and southern parts of the state, as brought out by Lindsey (1953), is much less obvious when the beech forest data are included in the analysis. The gap may not be bridged but the intermittent distance is greatly reduced.

SUMMARY

The present study was a part of the intensive investigation of Wisconsin plant communities being carried out at the University of Wisconsin. The primary intent of the phase of the study reported on here was to understand the phytosociology of the beech forests of the state and the relationships of these forests to other upland forests without beech.

The beech range in the state lies entirely within the limits of Wisconsin glaciation, being confined to the eastern edge of the state. The reasons for its restriction are not completely understood, but there is some evidence that it is limited to the southwest by an unfavorable moisture regime, and to the northwest by inadequate thermal conditions. The soils of the beech forest area are mostly of the grey-brown forest type, with a tendency toward a mor humus in the northern stands and a mull humus in the southern stands.

Previous studies involving the beech forests of the state were mostly descriptive and did not consider beech throughout its Wisconsin range. According to Camp (personal communication, 1953) the Wisconsin beeches are predominantly of the red beech gene mixture, but the southern white beech and the more northern gray beech have also entered in.

The random pairs method was used to sample the tree stratum, one meter square quadrats were

used for herbs, shrubs, and seedlings, and a point count was made of saplings. Importance values were computed for trees, frequency for herbs, and relative density for saplings.

In line with previous interpretations the beech stands were divided into southern and northern categories on the basis of composition. Throughout the Wisconsin beech range the dominance of sugar maple and beech is apparent, with sugar maple tending to be more important, especially in the southern stands. In the southern stands basswood, ironwood, red oak, red elm, and white ash are of secondary importance. In the northern stands hemlock is the only secondary species of consistent importance, yellow birch, white birch, and white pine having next highest average importance values but at the same time varying a great deal in importance from one stand to another. Sugar maple saplings usually have the highest relative density in a stand, although beech predominates in some cases, particularly where root sprouting is favored.

In order to analyze the relationship of these forests to the appropriate southern and northern non-beech upland forests of the state, the behavior of secondary species was plotted independently of and against a gradient of increasing importance of a climax unit (sugar maple or a summation of sugar maple and beech). Herb correlations, and soil and light data were also used for purposes of understanding these relationships. From the various lines of evidence it is concluded that respective portions of the beech forests can be woven into the terminal ends of the southern and northern forest continua (Curtis and McIntosh 1951, and Brown and Curtis 1952), and that in southern Wisconsin the Sugar Maple-Basswood and Beech-Sugar Maple are not distinct types. The inclusion of data from the beech forests also allows a closing of the gap between the forest areas of northern and southern Wisconsin.

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WHITE CEDAR ECOTYPES IN WISCONSIN¹

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INTRODUCTION

Within its natural range in Wisconsin, white cedar (*Thuja occidentalis* L.) displays the ability to persist in a wide range of habitats. Figure 1 indicates the range of white cedar in Wisconsin. It forms dense swamps on low, poorly drained topography and is commonly found on upland, well drained sites as well. This ability of white cedar has long been recognized and has been attributed to a wide degree of adaptability possessed by the species. Potzger (1941) has suggested that white cedar in northern Michigan possesses ecotypes because of its presence on upland and lowland sites in that area. There has not been, as yet, any critical investigation made of this phenomenon to substantiate the idea that ecotypes exist in white cedar. Taxonomically, there are no subspecies or varieties recognized in this species, there being no distinguishable morphological differences between white cedar found on any site.

This study was undertaken to determine whether ecotypic variants exist in the white cedar population in Wisconsin. In northern and eastern Wisconsin, the extremes in white cedar habitats can often be found geographically close together, sometimes within a few hundred feet. Wide geographical distances are involved in many demonstrations of ecotypes in other species. Pauley and Perry (1954) have shown that ecotypes occur in the wide ranging species of *Populus deltoides* Marsh. and *P. trichocarpa* Torr. & Gray in response to photoperiodicity. Vaartaja (1954) has determined the presence of photoperiodic ecotypes within *Pinus sylvestris* L. and *Alnus incana* (L.) Moench. as a result of studies on material collected from northern and southern latitudes in Finland. Wright (1944a, 1944b) found three ecotypes in both white ash (*Fraxinus americana* L.) and red ash (*F. pennsylvanica* Marsh.). These ash ecotypes are rather widely separated geographically and are differentiated on their ability to withstand cold.

¹Contribution from the University of Wisconsin Arboretum. Journal Paper No. 39.

Within a much smaller geographical area, central California, McMillan (1956) has indicated that important edaphic ecotypic variation exists in species of *Cupressus* and *Pinus* whose ranges include serpentine and nonserpentine soils. Ecotypes occurring rather close together on different soils are probably of greater ecological importance than those existing hundreds of miles apart on the same soil. The recognition of local ecotypes is of particular importance in phytosociological studies.

The procedure used in this study involved the collection, for transplantation purposes, of naturally occurring white cedar seedlings from both upland and lowland habitats throughout the white cedar range in Wisconsin, and also the collection of seed from the same sites for germination tests and growth experiments. A presentation of the methods and results will be given first for the seedling transplants, followed by a similar outline for the seed germination and growth experiments.

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WHITE CEDAR TRANSPLANTATION EXPERIMENT *Comparison of Collection Sites*

Figure 1 indicates the distribution of the seedling and seed collection sites. There is a total of twenty-nine sites, of which 16 are classified as typical lowland swamps consisting of nearly pure *Thuja occidentalis* with an admixture of balsam fir (*Abies balsamea*) and black ash (*Fraxinus nigra*); the remaining thirteen sites are well drained uplands, supporting in some cases pure white cedar, but often associated with hemlock (*Tsuga canadensis*), yellow birch (*Betula lutea*) and red maple (*Acer rubrum*). The lowland sites are frequently inundated during some part of the growing season, usually in spring, and tend to dry out in late summer, but still remain in a rather moist condition. The upland sites are also moist